

CSS Codes

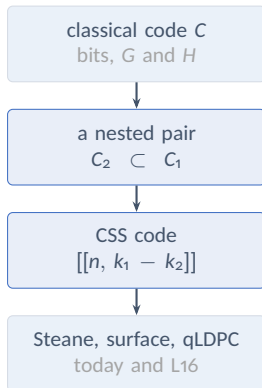
Building quantum codes from pairs of classical codes

Lecture 15

Term 261

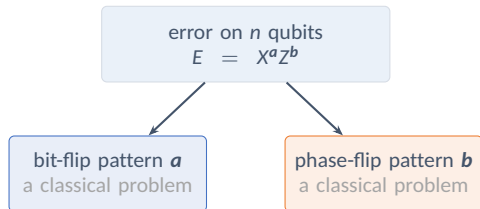
Agenda

- 1 Why go back to classical codes
- 2 Classical linear codes: G , H , syndromes, Hamming $[7, 4, 3]$, dual codes
- 3 The CSS construction: two classical codes, one quantum code
- 4 The Steane code $[[7, 1, 3]]$, decoded step by step
- 5 Transversal gates, and CSS codes everywhere



Why go back to classical codes?

- Shor's code spends **9 qubits** on one logical qubit: $R = k/n = 1/9$
- $[[5, 1, 3]]$ is the smallest single-error code (L13: $2(1 + 3 \cdot 5) = 2^5$), but checks like $XZZXI$ mix X and Z : awkward to decode, few easy gates
- Errors discretize: every Pauli error is, up to phase, $E = X^a Z^b$ with $\mathbf{a}, \mathbf{b} \in \mathbb{F}_2^n$
- $Y = iXZ$ sets a bit in *both* $\mathbf{a}$ and $\mathbf{b}$



e.g. $Y_5 = iX_5Z_5$: $\mathbf{a} = \mathbf{b} = 0000100$

Takeaway: Bit flips and phase flips are two separate classical decoding problems. Solve each with a classical code, and 75 years of classical coding theory (Hamming, 1950) comes for free.

Classical linear codes over $\mathbb{F}_2$

- An $[n, k]$ code C encodes k message bits into n bits; its elements are *codewords*
- **Linear:** $c, c' \in C \Rightarrow c \oplus c' \in C$. So C is a subspace of $\mathbb{F}_2^n$ and $|C| = 2^k$
- **Generator matrix** G , size $n \times k$, columns a basis of C :

$$C = \text{range}(G) = \{ G\mathbf{x} : \mathbf{x} \in \mathbb{F}_2^k \}$$

- **Distance:** $d(c, c') = \text{wt}(c \oplus c')$, so for a linear code $d =$ smallest weight of a nonzero codeword; write $[n, k, d]$

Example: $[3, 1, 3]$ repetition

$$G = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} : \quad G0 = 000, \quad G1 = 111$$

$$C = \{000, 111\}, \quad d = 3, \quad t = \lfloor \frac{d-1}{2} \rfloor = 1$$

All arithmetic is mod 2 ($1 \oplus 1 = 0$). Many coding texts (e.g. MacWilliams & Sloane) use the transposed row convention: G is $k \times n$ and codewords are $\mathbf{x}^T G$. We follow the T251 notes and Nielsen & Chuang.

The parity-check matrix

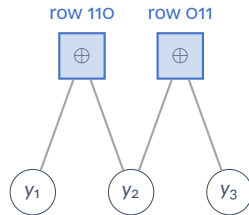
- **Parity-check matrix** H , size $(n - k) \times n$: each row is one parity check

$$C = \ker H = \{ \mathbf{y} \in \mathbb{F}_2^n : H\mathbf{y} = \mathbf{0} \}$$

- Every codeword passes every check, so $H\mathbf{G} = \mathbf{0}$
- G makes codewords; H tests them

Example: $[3, 1, 3]$

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad H \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = H\mathbf{G}$$



Tanner graph: bits below, checks above. The checks are Z_1Z_2 and Z_2Z_3 of the bit-flip code (L12).

Syndrome decoding

Send codeword c ; the channel adds an error e , we receive $y = c \oplus e$.

$$s = Hy = \underbrace{Hc}_{=0} \oplus He = He$$

- The **syndrome** depends only on the error, not on the message
- Decoder: look up the most likely (lowest-weight) e with $He = s$, output $y \oplus e$
- 2^{n-k} syndromes, so at most 2^{n-k} errors can be told apart

[3, 1, 3] syndrome table

error e	$s = He$	(Z_1Z_2, Z_2Z_3)	fix
000	00	$(+1, +1)$	none
100	10	$(-1, +1)$	bit 1
010	11	$(-1, -1)$	bit 2
001	01	$(+1, -1)$	bit 3

Syndrome bit 1 $\leftrightarrow$ eigenvalue -1 : the quantum bit-flip code of L12 is this table. At $p = 0.01$:

$$p_L = 3p^2 - 2p^3 = 2.98 \times 10^{-4}.$$

The Hamming [7, 4, 3] code

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \quad H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

G is 7×4 , H is 3×7 , $HG = 0 \pmod{2}$ — checked

- $2^4 = 16$ codewords; weights: one of weight 0, seven of weight 3, seven of weight 4, one of weight 7
- So $d = 3$, $t = 1$
- Rate $R = 4/7 \approx 0.57$, versus $1/3$ for $[3, 1, 3]$ at the same t

Encode $x = 1011$

Top block of G is I_4 , so the message is copied:

$$Gx = 1011 \mid 010$$

three parity bits appended.

The T251 notes called $n/k = 7/4 = 1.75$ the “rate”; we call n/k the *overhead* and keep $R = k/n$ (L13).

Hamming syndromes spell the error position

Column j of H is j written in binary:

bit j	1	2	3	4	5	6	7
row 1 (4s)	0	0	0	1	1	1	1
row 2 (2s)	0	1	1	0	0	1	1
row 3 (1s)	1	0	1	0	1	0	1

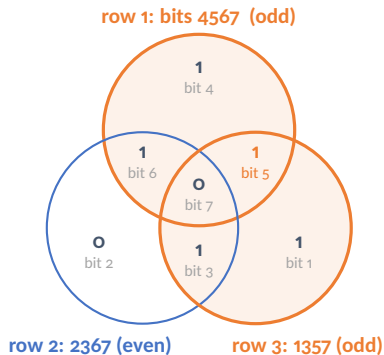
A flip on bit j gives $He_j = \text{column } j = j \text{ in binary.}$

Worked

$c = 1011010$; flip bit 5: $y = 1011110$

$s = Hy = (1, 0, 1) = 5 \Rightarrow \text{flip bit 5 back}$

All $16 \times 7 = 112$ single flips decoded correctly.



Each circle must hold an even number of 1s. Bit 5 is the only bit inside both odd circles and outside the even one.

The dual code

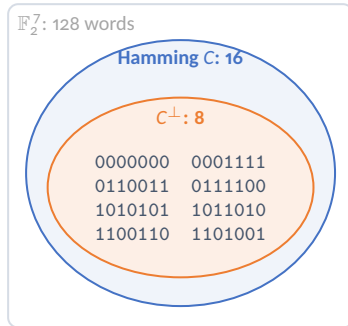
$$C^\perp = \{ \mathbf{y} \in \mathbb{F}_2^n : \mathbf{y} \cdot \mathbf{x} = 0 \ \forall \mathbf{x} \in C \}$$

- If C is $[n, k]$ with (G, H) , then $C^\perp$ is $[n, n - k]$ with generator H^T and parity check G^T
- Over $\mathbb{F}_2$ a word can be orthogonal to itself ($1100 \cdot 1100 = 0$), so C and $C^\perp$ can overlap

Hamming dual: the simplex $[7,3,4]$ code

All 7 nonzero words have weight 4, and $C^\perp \subset C$: they are exactly the **even-weight** Hamming codewords.

Is $C^\perp \subset C$ for the repetition code $[3, 1, 3]$? (No: $C^\perp = \{000, 110, 011, 101\}$.)



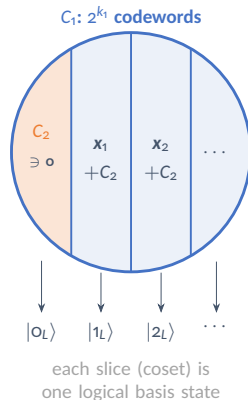
The CSS construction

Take classical codes $C_1 = [n, k_1]$ and $C_2 = [n, k_2]$ with $C_2 \subset C_1$. For $\mathbf{x} \in C_1$:

$$|\mathbf{x} + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{\mathbf{y} \in C_2} |\mathbf{x} \oplus \mathbf{y}\rangle$$

- Same coset ($\mathbf{x} \oplus \mathbf{x}' \in C_2$) $\Rightarrow$ same state
- Different cosets are disjoint sets $\Rightarrow$ orthogonal states
- $|C_1|/|C_2| = 2^{k_1-k_2}$ cosets $\Rightarrow 2^{k_1-k_2}$ basis states

Result: $\text{CSS}(C_1, C_2)$ is an $[[n, k_1 - k_2]]$ quantum code.

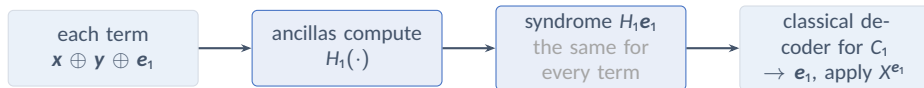


What is $\text{CSS}(C_1, C_2)$ for $C_1 = \mathbb{F}_2^n$, $C_2 = \{\mathbf{o}\}$? ($[[n, n]]$: every state is a codeword, nothing is protected.)

Bit flips: a classical problem for C_1

An X error X^{e_1} adds e_1 to every term of the superposition:

$$X^{e_1} |\mathbf{x} + C_2\rangle = \frac{1}{\sqrt{|C_2|}} \sum_{\mathbf{y} \in C_2} |\mathbf{x} \oplus \mathbf{y} \oplus \mathbf{e}_1\rangle$$



- $\mathbf{x} \oplus \mathbf{y} \in C_1$, so $H_1(\mathbf{x} \oplus \mathbf{y} \oplus \mathbf{e}_1) = H_1\mathbf{e}_1$: the message and the branch $\mathbf{y}$ drop out
- Every branch gives the same syndrome, so measuring it does **not** collapse the superposition
- Corrects up to t bit flips if $d(C_1) \geq 2t + 1$

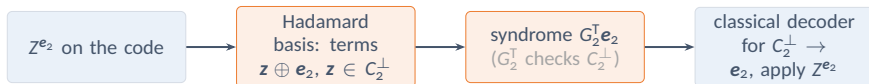
Takeaway: In the computational basis, a CSS code is just C_1 protecting against bit flips.

Phase flips: a classical problem for $C_2^\perp$

Apply $H^{\otimes n}$. Since $H^{\otimes n} |y\rangle = 2^{-n/2} \sum_z (-1)^{y \cdot z} |z\rangle$ and $\sum_{y \in C_2} (-1)^{y \cdot z}$ vanishes unless $z \in C_2^\perp$:

$$H^{\otimes n} |x + C_2\rangle = \frac{1}{\sqrt{|C_2^\perp|}} \sum_{z \in C_2^\perp} (-1)^{x \cdot z} |z\rangle$$

And $HZH = X$: a phase error Z^{e_2} becomes a bit flip X^{e_2} on a superposition over $C_2^\perp$.



CSS error-correction guarantee (Calderbank-Shor, Steane 1996)

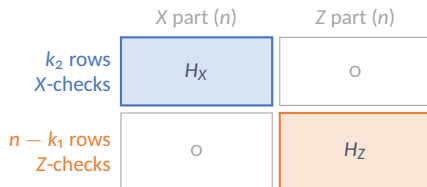
If $d(C_1) \geq 2t + 1$ and $d(C_2^\perp) \geq 2t + 1$, the code corrects any t bit flips *and* any t phase flips — hence any error on t qubits.

In practice no $H^{\otimes n}$ is applied: measuring the X-type checks below does the same job.

CSS codes in stabilizer language (L14)

- **Z-checks:** one Z^h per row $\mathbf{h}$ of $H_Z = H_1$.
 Z^h anticommutes with X^e iff $\mathbf{h} \cdot \mathbf{e} = 1 \Rightarrow$
syndrome $H_1 \mathbf{e}_1$
- **X-checks:** one X^g per row $\mathbf{g}$ of $H_X = G_2^T$ (the
parity check of $C_2^\perp$) $\Rightarrow$ syndrome $G_2^T \mathbf{e}_2$
- **They commute:** X^g, Z^h commute iff $\mathbf{g} \cdot \mathbf{h} = 0$,
and $H_X H_Z^T = (H_1 G_2)^T = 0$ since every column
of G_2 lies in $C_2 \subset C_1 = \ker H_1$

Check matrix ($x \mid z$)



Every generator is purely X- or purely Z-type.
Logical qubits:

$$k = n - \text{rk } H_X - \text{rk } H_Z = k_1 - k_2$$

Takeaway: A CSS code is a stabilizer code whose check matrix is block-diagonal.

The Steane code $[[7, 1, 3]]$

- $C_1 = \text{Hamming } [7, 4, 3]$, $C_2 = C_1^\perp = \text{simplex } [7, 3, 4]$; $C_2 \subset C_1$ ✓
- $C_2^\perp = C_1$, so $H_X = H_Z = H$: the X - and Z -checks have **the same pattern**
- $k = 4 - 3 = 1$; $d(C_1) = d(C_2^\perp) = 3 \Rightarrow$ corrects any single-qubit error
- $R = 1/7 \approx 0.14$, versus $1/9 \approx 0.11$ for Shor

$$\left(\begin{array}{c|c} H & O \\ \hline O & H \end{array} \right) \text{ with } H = \begin{pmatrix} 0001111 \\ 0110011 \\ 1010101 \end{pmatrix}$$

The six stabilizer generators

from row	X-type	Z-type
0001111	IIIXXXX	IIIZZZZ
0110011	IXXIIXX	IZZIIZZ
1010101	XIXIXIX	ZIZIZIZ

Every pair commutes (checked): any X -row and Z -row overlap on an even number of qubits (2 or 4).

Six weight-4 checks on 7 qubits leave $2^{7-6} = 2$ dimensions: one logical qubit.

Steane codewords and logical operators

$$|o_L\rangle = |o + C_2\rangle = \frac{1}{\sqrt{8}} \sum_{y \in C_1^\perp} |y\rangle$$

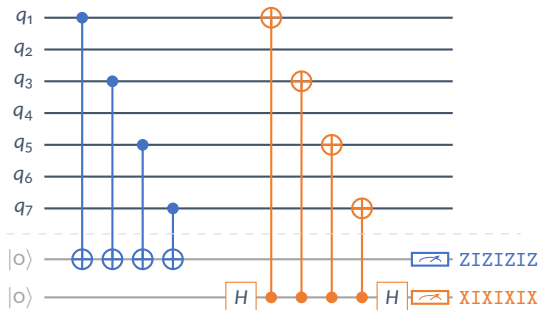
$$|1_L\rangle = |1111111 + C_2\rangle = \bar{X} |o_L\rangle$$

- $\bar{X} = X^{\otimes 7}$, $\bar{Z} = Z^{\otimes 7}$: $\bar{Z} |o_L\rangle = |o_L\rangle$, $\bar{Z} |1_L\rangle = -|1_L\rangle$
(even vs odd weight)
- Multiply by stabilizers to shorten them: $\bar{X} \simeq X_1 X_2 X_3$,
 $\bar{Z} \simeq Z_1 Z_2 Z_3$ (1110000 is a Hamming codeword not in C_2)
- No logical operator has weight ≤ 2 (brute force) $\Rightarrow$
 $d = 3$

$ o_L\rangle$: even weight	$ 1_L\rangle$: odd weight
0000000	1111111
0001111	1110000
0110011	1001100
0111100	1000011
1010101	0101010
1011010	0100101
1100110	0011001
1101001	0010110

Each row: right = left $\oplus$ 1111111.
Together: all 16 Hamming codewords.

Measuring the Steane checks



- **Z-check**: the ancilla collects the parity of qubits 1,3,5,7 (as in L12)
- **X-check**: the same circuit in the Hadamard basis; CNOTs point the other way
- One ancilla per row: 6 ancillas, $6 \times 4 = 24$ CNOTs per round
- Outcome 1 $\leftrightarrow$ eigenvalue -1

Two of the six checks shown. Making extraction itself tolerant of faulty ancillas is a fault-tolerance topic.

Worked decoding: a Y error on qubit 5

Encode a random state $|\psi\rangle = \alpha |0_L\rangle + \beta |1_L\rangle$, $\alpha = -0.705 + 0.604i$, $\beta = 0.322 + 0.187i$ (128-dim statevector). Apply $Y_5 = iX_5Z_5$.



X-checks (see the Z part)

check	overlaps 5?	outcome
IIIXXXX	yes	$-1 \rightarrow \mathbf{1}$
IXXIIXX	no	$+1 \rightarrow \mathbf{0}$
XIXIXIX	yes	$-1 \rightarrow \mathbf{1}$

syndrome 101 = 5 $\Rightarrow$ Z on qubit 5

Z-checks (see the X part)

check	overlaps 5?	outcome
IIIZZZZ	yes	$-1 \rightarrow \mathbf{1}$
IZZIIZZ	no	$+1 \rightarrow \mathbf{0}$
ZIZIZIZ	yes	$-1 \rightarrow \mathbf{1}$

syndrome 101 = 5 $\Rightarrow$ X on qubit 5

Result

Both Hamming decoders point at column 5 = (1, 0, 1). Apply Y_5 : fidelity with $|\psi\rangle$ goes from 0 (after the error) to 1.000. All 21 single-qubit X, Y, Z errors are corrected the same way.

Two errors, two decoders: X_2 and Z_6 together

Same $|\psi\rangle$; now the error is X_2Z_6 — weight 2, more than $t = 1$.



Z-checks read 010 = 2
 $\Rightarrow$ X error on qubit 2

X-checks read 110 = 6
 $\Rightarrow$ Z error on qubit 6

Apply X_2Z_6 : fidelity 1.000. All $7 \times 7 = 49$ combinations X_iZ_j are corrected.

Why this works

The two decoders never talk to each other: one weight-1 X error *plus* one weight-1 Z error is within reach of both Hamming codes.

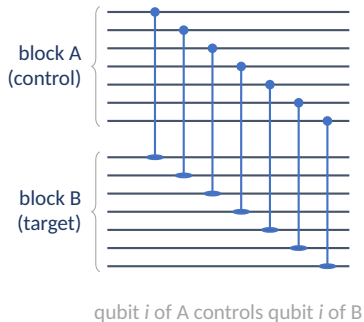
The catch: two errors of the *same* type fail. X_1X_2 : Z-syndrome 011 = 3, the decoder applies X_3 , and $X_1X_2X_3 = \bar{X}$ — a silent logical error.

Transversal gates on the Steane code

Transversal: the same gate on every qubit (or on qubit i of each block). Each line below was checked on the code space, with zero leakage out of it:

physical	logical	why
$H^{\otimes 7}$	$\bar{H}$	swaps X- and Z-checks: same pattern
$S^{\otimes 7}$	$\bar{S}^\dagger$	phase i^{wt} : wt 0, 4 $\rightarrow$ 1; wt 3, 7 $\rightarrow$ $-i$
$\text{CNOT}^{\otimes 7}$	$\overline{\text{CNOT}}$	cosets add: $\mathbf{x}' + C_2 \rightarrow \mathbf{x} \oplus \mathbf{x}' + C_2$

Together with $\bar{X} = X^{\otimes 7}$, $\bar{Z} = Z^{\otimes 7}$: H , S and CNOT generate the Clifford group, so the Steane code has **every Clifford gate transversally** ($\bar{S} = (S^\dagger)^{\otimes 7}$).

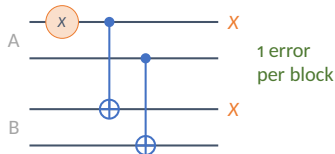


Why transversal means fault-tolerant

gate inside one block



transversal gate



- A transversal gate never couples two qubits of the same block
- One faulty gate leaves at most one error per block — which the code corrects

Eastin-Knill (2009)

No code that detects every single-qubit error has a *universal* set of transversal gates.

Steane: all Cliffords, no T . $[[15, 1, 3]]$ Reed-Muller: $T^{\otimes 15} = \bar{T}^\dagger$, but no transversal H . The missing gate needs extra machinery (magic states).

B. Eastin, E. Knill, PRL 102, 110502 (2009); S. Bravyi, A. Kitaev, PRA 71, 022316 (2005).

CSS codes are everywhere

Code	$[[n, k, d]]$	Built from	Check wt	R	Known for
Shor (1995)	$[[9, 1, 3]]$	repetition codes	Z: 2, X: 6	0.11	first quantum code (L12)
Steane	$[[7, 1, 3]]$	Hamming $\supset$ simplex	4	0.14	transversal Cliffords
Reed-Muller	$[[15, 1, 3]]$	punctured Reed-Muller	4, 8	0.07	transversal T
Surface	$[[d^2, 1, d]]$	2D grid (L16)	4, local	$\rightarrow 0$	nearest-neighbour checks
Gross (BB)	$[[144, 12, 12]]$	cyclic shifts (L16)	6	0.083	qLDPC, high rate

Rows 1–4: $H_X H_Z^T = 0$, k and d checked by brute force (surface code at $d = 3$: $[[9, 1, 3]]$).

Takeaway: L16 builds the surface code and qLDPC codes (hypergraph products, bivariate bicycle). All are CSS: sparse H_X, H_Z with $H_X H_Z^T = 0$ and two classical decoders, exactly as today.

Gross code: S. Bravyi et al., Nature 627, 778 (2024). Hypergraph product: Tillich & Zémor, IEEE Trans. Inf. Theory 60, 1193 (2014).

Summary

- A classical $[n, k, d]$ linear code: $C = \{G\mathbf{x}\} = \ker H$, $HG = 0$; the syndrome $H\mathbf{y} = H\mathbf{e}$ sees only the error
- Hamming $[7, 4, 3]$: the syndrome is the error position in binary; its dual is the simplex $[7, 3, 4]$ code, and $C^\perp \subset C$
- CSS: $C_2 \subset C_1$ gives $[[n, k_1 - k_2]]$ with codewords $|\mathbf{x} + C_2\rangle$; X errors are decoded with C_1 , Z errors with $C_2^\perp$
- Stabilizer view: $H_Z = H_1$, $H_X = G_2^\top$, and $H_X H_Z^\top = 0$ makes them commute
- Steane $[[7, 1, 3]]$: six weight-4 checks; Y_5 gives syndrome 101 twice; $X_2 Z_6$ is fixed by two independent Hamming decoders
- Transversal H , $S (\rightarrow \bar{S}^\dagger)$ and CNOT give all Cliffords; Eastin–Knill says no code gets a universal transversal set
- Surface and qLDPC codes (L16) are CSS codes with sparse, structured H_X and H_Z

References

CSS codes and the Steane code

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Classical coding theory

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Transversal gates, and CSS codes beyond Steane

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All matrices, codewords, syndromes and gate actions on these slides were computed in scripts/L15_classical.py, L15_steane.py and L15_css_table.py.